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Unmanned Powered Paraglider Flight Path Control Based on PID Neutral Network

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Unmanned Powered Paraglider Flight Path Control Based on PID Neutral Network

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Abstract. The unmanned powered paraglider (UPP) is a nonlinear time-varying dynamic system with high coupling degree. In view of how to make the UPP flying along arbitrary predetermined trajectory, the conventional control method is difficult to obtain the control trajectory with a better fitting degree of the presupposition trajectory. To solve this problem, the six degree of freedom (SDOF)dynamic model of the UPP is built and the control characteristics of the UPP is studied. In view of the control characteristics of the UPP, the controller suitable for the UPP is designed and compiled with the PID neural network method (PIDNN), and the simulation system for the flight process is built, which is used to simulate the flight path control. And the results show that the controller realizes the effective flight path control of the UPP.

1. Introduction

The unmanned powered paraglider (UPP) is a kind of flexible wing aircraft, which provides the lift needed by the parachute. It is mainly composed of four parts: parachute, control rope, propulsion system and payload cabin. Compared with other aircraft, the UPP has the advantages of lower cost, higher load capacity and less requirement for the site.[1][2]

However, due to the low flying speed of the UPP and the relatively high uncertainty of the complex flight environment, the magnitude of the disturbance is close to the flight speed, which makes it easier to be affected. In addition, because of the high coupling degree, nonlinear and time-varying dynamic characteristics of the UPP, the input control amount are only the control rope and the propeller thrust, and the dimension of the input is smaller when compared with the output. It also makes the self-adaptive of the control algorithm very necessary.[3]

In view of the above problems, to realize the autonomous flight according to a certain route, a more stable, reliable and adaptive control method is needed for the UPP. The PID neural network (PIDNN) method inherits the advantages of PID’s simple structure and high reliability, and it involve the neural network algorithm. Although the traditional PID has simple structure and strong reliability [4][5], it is difficult to obtain good control result for the dynamic system with high coupling and nonlinear time-varying characteristics. To improve its self-adaptability[6][7], we consider the combination of neural network and PID control. There are two ways to combine the traditional neural network with the PID control. One is to determine the PID parameters by using the neural network.[8]-[10] Its structure is much more complex than the traditional PID controller and cannot avoid the weakness of slow convergence speed of the general neural network. It is also difficult to determine the number of neurons in hidden layer and the weight value between different layers. The other one is a single neuron structure PID controller[11], but it does not have the ability to approximate any function, so it is difficult to achieve good result in the complex control system such as UPP.

The PID neural network in this paper, instead of combining neural network and traditional PID control law, defines neurons with proportional, integral and differential functions, and uses neural network to adjust the connection weight directly. It inherits the advantages of traditional PID and neural network and avoids the shortcomings of the general method of combining PID with neural network.[12][13]

In this paper, the six degree of freedom (SDOF) model of the UPP is established. And the control characteristics is analysed based on the model. Then, the PID neural network method is used to design the controller for this system. Finally, according to the result of simulation, the UPP's flight path is effectively in accordance with the pre-set trajectory.

2. Six degree of freedom modeling (SDOF) of UPP

The SDOF model of the UPP is the basis of investigating the control problem. Considering the characteristics of the parafoil part, the force analysis requires the establishment of additional mass. And considering that the actual shape of the parafoil is curved, it should be differentiated into several same small parts in which the aerodynamic force and the aerodynamic torque should be considered.

2.1. The basic hypothesis and the selection of coordinate system

Some basic assumptions are as follows. First, the connections between parafoil and the load tank are considered as rigid connections, and the mass and the moment of inertia of the parafoil rope can be neglected. Second, the load compartment is regarded as a standard square with uniform mass distribution, and its lift and aerodynamic force, aerodynamic moment can be neglected, only its resistance is considered. Third: the quality center and pressure center of the parafoil are located at the 1/4 distance from the leading edge. Fourth: the hypothesis of the plane mechanism for earth.

Three coordinates are needed, which are the body coordinate system S_p, the geodetic coordinate system S_g and the air flow coordinate system S_a. The origin of S_g is O_g located at the center of mass of the model at the initial time. And the axis O_gZ_g points vertically to the earth. The O_gX_g points toward the leading edge along the horizontal direction, and O_gY_g forms the right-hand line with other two axes. As for the S_a, the origin O_a is located at the quarter of the string, and the O_aX_a is to the opposite direction of the large air flow. The O_aZ_a is in the vertical symmetry plane of the parachute, which is vertical to the O_aX_a, and points toward the lower wing surface of the wing. Similarly, the O_aY_a forms the right-hand line with other two axes. The three coordinate systems are shown in Figure 1 as follows:

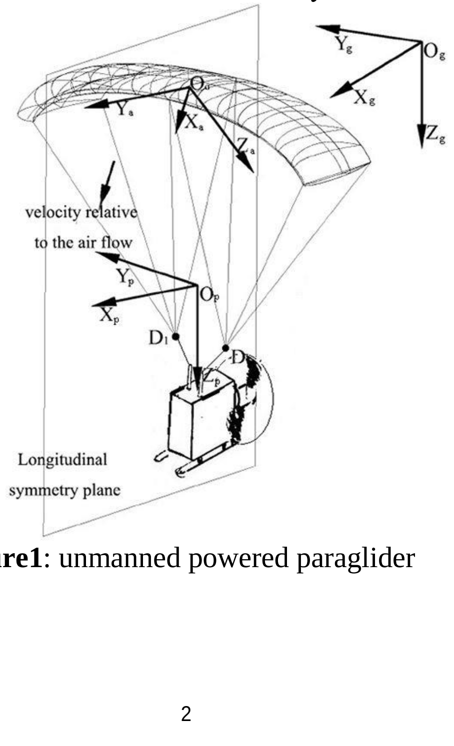


Figure1: unmanned powered paraglider

2.2. The force analysis and the establishment of dynamic equation

The force analysis of the UPP contains the force analysis of the parafoil and of the loading cabin. For the record, additional mass is needed and the force of the parafoil should be set up in different same section because of the inflatable structure and arc structure of the parafoil.

2.2.1. Additional mass and additional moment of inertia. The parafoil of UPP is inflated with air during the flight. Therefore, the density of the parafoil part of the UPP is similar to that of the surrounding fluid. The effect of the surrounding fluid on the whole motion can be added to the total mass and the moment of inertia in the form of additional mass and additional moment of inertia. The additional mass matrix is established by the method of Barrows 6 x 6 additional matrix[14], and the additional mass and additional moment of inertia is obtained as follows:

$$\boldsymbol{A}_{\mathrm{a,O}} = \begin{bmatrix} \boldsymbol{M}_{\mathrm{a}} & -\boldsymbol{M}_{\mathrm{a}}(\boldsymbol{L}_{\mathrm{O\text{-}R}}^{\times} + \boldsymbol{L}_{\mathrm{R\text{-}P}}^{\times}\boldsymbol{B}_2) \\ (\boldsymbol{B}_2\boldsymbol{L}_{\mathrm{R\text{-}P}}^{\times} + \boldsymbol{L}_{\mathrm{O\text{-}R}}^{\times})\boldsymbol{M}_{\mathrm{a}} & \boldsymbol{J}_{\mathrm{a,O}} \end{bmatrix}, \quad \boldsymbol{B}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{1)(2}$$

The internal terms in the upper matrix are as follows:

$$\boldsymbol{J}_{\mathrm{a,O}} = \boldsymbol{J}_{\mathrm{a}} - \boldsymbol{L}_{\mathrm{O\text{-}R}}^{\times}\boldsymbol{M}_{\mathrm{a}}\boldsymbol{L}_{\mathrm{O\text{-}R}}^{\times} - \boldsymbol{L}_{\mathrm{R\text{-}P}}^{\times}\boldsymbol{M}_{\mathrm{a}}\boldsymbol{L}_{\mathrm{R\text{-}P}}^{\times}\boldsymbol{B}_2 - \boldsymbol{Q} - \boldsymbol{Q}^{\mathrm{T}} \tag{3}$$

$$\boldsymbol{Q} = \boldsymbol{B}_2\boldsymbol{L}_{\mathrm{R\text{-}P}}^{\times}\boldsymbol{M}_{\mathrm{a}}\boldsymbol{L}_{\mathrm{O\text{-}R}}^{\times} \tag{4}$$

$\boldsymbol{L}_{\mathrm{O\text{-}R}}^{\times}$ and $\boldsymbol{L}_{\mathrm{R\text{-}P}}^{\times}$ are respectively the multiplication matrices of the vector pointing to the parafoil's roll center from the center of mass and the vector pointing from the parafoil's pitching center from its roll center. $\boldsymbol{M}_{\mathrm{a}}$ and $\boldsymbol{J}_{\mathrm{a}}$ are diagonal matrixes of which diagonal elements are the additional mass and the additional moment of inertia for the parafoil.

2.2.2. Aerodynamic force and aerodynamic torque. Considering the arc structure of the parachute, the parafoil is divided into eight segments on average, and the respective body coordinate system is set up in each part, and the lift and resistance are obtained as follows:

$$\boldsymbol{F}_{\mathrm{L}_i} = \frac{1}{2}\rho S_i\sqrt{u_{\mathrm{p}i}^2 + w_{\mathrm{p}i}^2}\,k_i C_{\mathrm{L}_i}\begin{bmatrix} w_{\mathrm{p}i} \\ 0 \\ -u_{\mathrm{p}i} \end{bmatrix} \qquad \boldsymbol{F}_{\mathrm{D}_i} = -\frac{1}{2}\rho S_i V_{\mathrm{p}i} C_{\mathrm{D}_i}\begin{bmatrix} u_{\mathrm{p}i} \\ v_{\mathrm{p}i} \\ w_{\mathrm{p}i} \end{bmatrix} \tag{5)(6}$$

The subscript i represents the physical quantity corresponding to the section i; k is a product factor. S is the area of each segment of pterygoid; u_p, v_p and w_p are the projection of the velocity of the parafoil's each segment in respective coordinate system; V_p is the resultant velocity; C_L, C_D is the lift and drag coefficient. The total aerodynamics and aerodynamic torque expressions of the parafoil can be obtained by converting those of different section to the body coordinate system of the UPP.

$$\boldsymbol{F}_{\mathrm{aerop}} = \sum_{i=1}^{8}\boldsymbol{B}_{i\text{-O}}(\boldsymbol{F}_{\mathrm{L}_i} + \boldsymbol{F}_{\mathrm{D}_i}) \qquad \boldsymbol{M}_{\mathrm{aerop}} = \sum_{i=1}^{8}\boldsymbol{L}_{\mathrm{O}-i}^{\times}\boldsymbol{B}_{i\text{-O}}(\boldsymbol{F}_{\mathrm{L}_i} + \boldsymbol{F}_{\mathrm{D}_i}) \tag{7)(8}$$

As for the the load tank, because it does not produce the lift, it is only necessary to analyse its drag. So, its aerodynamic force is as follows:

$$\boldsymbol{F}_{\mathrm{aerob}} = -\frac{1}{2}\rho S_{\mathrm{b}} V_{\mathrm{b}} C_{\mathrm{db}}\begin{bmatrix} u_{\mathrm{b}} \\ v_{\mathrm{b}} \\ w_{\mathrm{b}} \end{bmatrix} \tag{9}$$

Sb is the section area of the load tank. C_{db} is the drag coefficient. V_b is the resultant velocity of the load tank. u_b, v_b and w_b are the projection of the load tank's velocity in body coordinate system.

2.2.3. Dynamic model. Arranging the dynamic equations of the UPP in matrix form, the dynamic equations are as follows. G is the gravity component array. S_w is the multiplication matrix of the angular velocity, and Ft is the thrust provided by power system.

$$\begin{bmatrix} m_r \boldsymbol{I} + \boldsymbol{M}_a & -\boldsymbol{M}_a (\boldsymbol{L}_{\text{O-R}}^{\times} + \boldsymbol{L}_{\text{R-P}}^{\times} \boldsymbol{B}_2) \\ (\boldsymbol{B}_2 \boldsymbol{L}_{\text{R-P}}^{\times} + \boldsymbol{L}_{\text{O-R}}^{\times}) \boldsymbol{M}_a & \boldsymbol{J}_r + \boldsymbol{J}_{a,O} \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{V}}_O \\ \dot{\boldsymbol{W}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{F}_{\text{total}} \\ \boldsymbol{M}_{\text{total}} \end{bmatrix} \tag{10}$$

The internal terms in the upper matrix are as follows. In the equation, the moment of inertia of the parachute to the centroid of the UPP is J_p, the moment of inertia of the load tank to the centroid of the system is J_b:

$$\boldsymbol{F}_{\text{total}} = \boldsymbol{F}_{\text{aerop}} + \boldsymbol{F}_{\text{aerob}} + \boldsymbol{F}_t + \boldsymbol{G} - \boldsymbol{S}_w^{\times} m_r \boldsymbol{V}_O - \boldsymbol{S}_w^{\times} \boldsymbol{M}_a \boldsymbol{V}_O + \boldsymbol{S}_w^{\times} \boldsymbol{M}_a \boldsymbol{D} \boldsymbol{W} \tag{11}$$

$$\begin{aligned} \boldsymbol{M}_{\text{total}} &= \boldsymbol{M}_{\text{aerop}} + \boldsymbol{L}_{\text{O-p}}^{\times} \boldsymbol{F}_{\text{aerop}} + \boldsymbol{L}_{\text{O-b}}^{\times} (\boldsymbol{F}_{\text{aerob}} + \boldsymbol{F}_t) - \boldsymbol{S}_w^{\times} (\boldsymbol{J}_r + \boldsymbol{J}_{a,O}) \boldsymbol{W} \\ &- \boldsymbol{S}_w^{\times} \boldsymbol{D}^T \boldsymbol{M}_a \boldsymbol{V}_O + \boldsymbol{S}_v^{\times} \boldsymbol{M}_a \boldsymbol{D} \boldsymbol{W} \end{aligned} \tag{12}$$

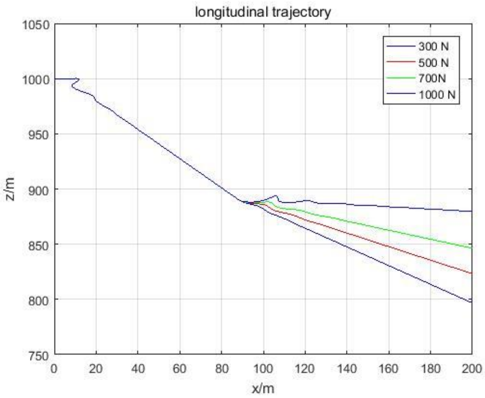
$$\boldsymbol{D} = \boldsymbol{L}_{\text{O-R}}^{\times} + \boldsymbol{L}_{\text{R-P}}^{\times} \boldsymbol{B}_2 \qquad \boldsymbol{J}_r = \boldsymbol{J}_p + \boldsymbol{J}_b \qquad (13)(14)$$

3. Control characteristic of SDOF model of UPP

The flight path control of the UPP is mainly divided into longitudinal control and yaw control. The longitudinal control refers to controlling the flight speed and the flight height of the aircraft through the change of the propeller thrust. The yaw control is mainly through asymmetrically pulling down control rope, which causes the asymmetrical deflection of the rear edge of the parafoil, thus generating the torque to control the direction of the flight.

3.1. Longitudinal control

When given the different thrust, the UPP has different trajectories in the vertical plane which are shown as follows:



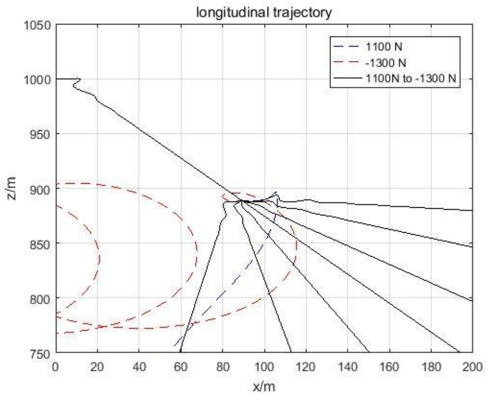


Figure2: longitude trajectory under different thrusts **Figure3**: Critical thrust to ensure no stall

It is known from Figure 2 that the different thrust is applied during the flight, and the trajectory is kept in the vertical plane because the device providing thrust is installed at the rear of the load tank, so there is no velocity component in the direction of the Y axis. After given thrust, the gliding ratio is improved. The difference is that the greater the thrust, the greater the glide ratio.

Besides, to prevent stall in the flight, the value of the thrust should be restrained in an effective range. According to the Figure 3, when the thrust is between 1100N and - 1300N, the flight trajectory of the UPP can be changed normally by the thrust. When the thrust exceeds this range, the stall happens. Therefore, when designing controller, the control volume should be controlled within this range.

3.2. Yaw control

When applied different pulling amount to the left control rope and the right control rope, the rear edge of the parachute produces an angle, which makes the rear edge of the parafoil asymmetrically deflected, and the trajectory response of the UPP based on the asymmetrical deflection of the following edge is shown as Figure 4.

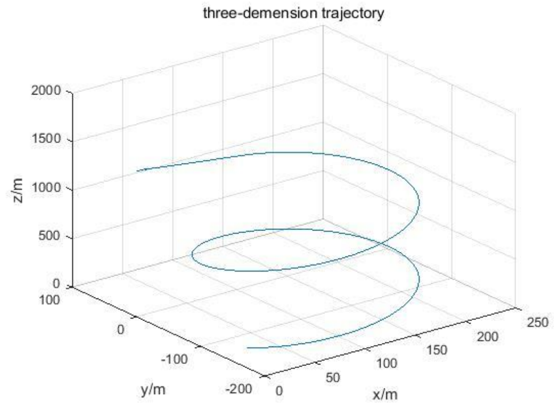


Figure 4: trajectory under the controlling of ropes

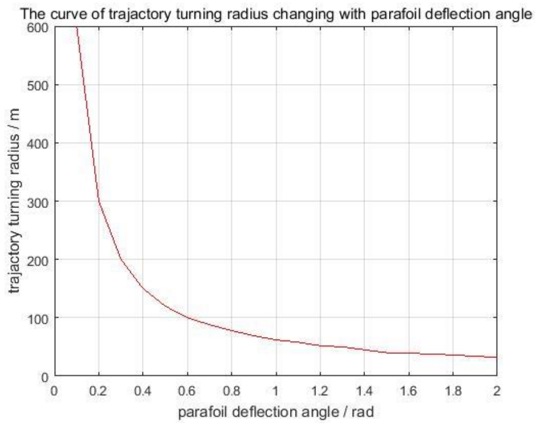


Figure 5: turning radius changing with parafoil deflection angle

It can be seen from Figure 4 that when the left control rope is pulled, the UPP begins to turn. Because of the difference of the parafoil's deflection angle created by the asymmetrical pulling amount of the left and right control ropes, the turning radius of the trajectory also changes. The relationship between these two variables can be obtained as Figure 5. It can be concluded that the turning radius of the UPP's trajectory decreases with the increase of the parafoil's rare edge deflection angle. And the minimum radius of the turning radius of the trajectory is about 30 meters. Therefore, the pre-set turning radius of the trajectory should not be set lower than this minimum according to the actual situation, otherwise the corresponding trajectory control cannot be achieved.

4. PID neutral network (PIDNN) structure and controller design

PID neural network (PIDNN) is not a simple combination of neural network and traditional PID control law. Instead of using neural network to set PID parameters, it defines the neuron with proportional, integral and differential function, and uses neural network to adjust the weight of connection directly[15].The most basic form of PIDNN is Single PID neural network (SPIDNN), which is a three level forward neural network of $2\times3\times1$. The input layer contains two neurons and receives external input information, which is the output feedback value and the pre-set value of the controlled object respectively. There are three neurons in the hidden layer, which have the function of the proportion, integral and differential operation respectively. The output layer has one neuron, which completes the synthesis of the outputs of hidden layer's neurons and then outputs the corresponding control volume. As is shown in Figure 6.

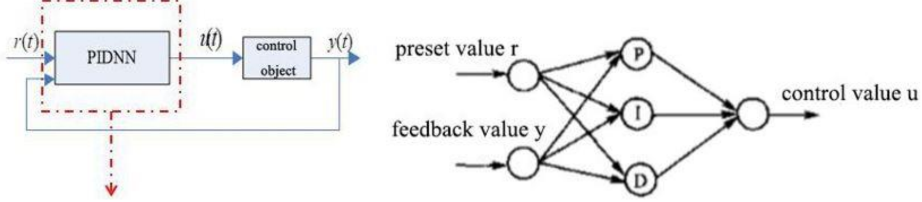


Figure 6: the most basic form of PIDNN

4.1. Network structure design

When the controlled object has m inputs and n outputs, MPIDNN needs 2n input units in the input layer, and there are 3n processing units in the hidden layer, including n proportional elements, n differential elements, n integral elements and there are m output units in the output layer. In the control process, the comparison between the actual position and the pre-set position is based on the position in the X direction, and the position in the Y direction and the Z direction is fed back to the controller. Figure 7 shows the control loop for UPP.

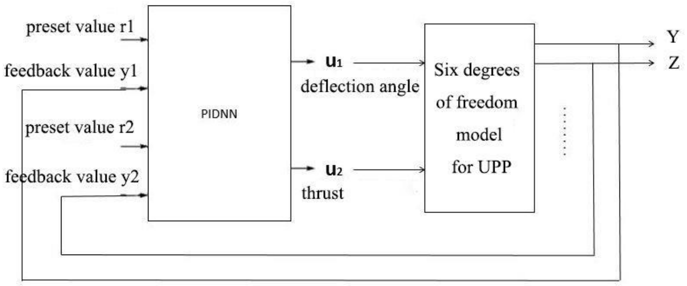


Figure 7: control loop for UPP

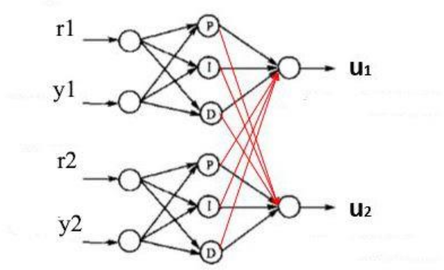


Figure 8: the structure of MPIDNN

Therefore, the structure of the PIDNN controller which is shown in Figure 8 is a 4×6×2 three-layer forward neuron network, including two basic SPIDNN structures, the input amounts are the Y direction pre-set value and feedback value, the Z direction pre-set value and feedback value. The output amounts are obtained through the forward algorithm of PIDNN. The outputs are the values of the two control variables received by the UPP. PIDNN controller then adjusts the network weights between different layers according to the error value and prepares for the next calculation.

4.2. Design of control algorithm

The control algorithm is composed of two parts: forward algorithm and backpropagation algorithm. The forward algorithm is used to calculating the value of the control variables, and the backpropagation algorithm is used to adjust the network weights. The process is shown in Figure 9:

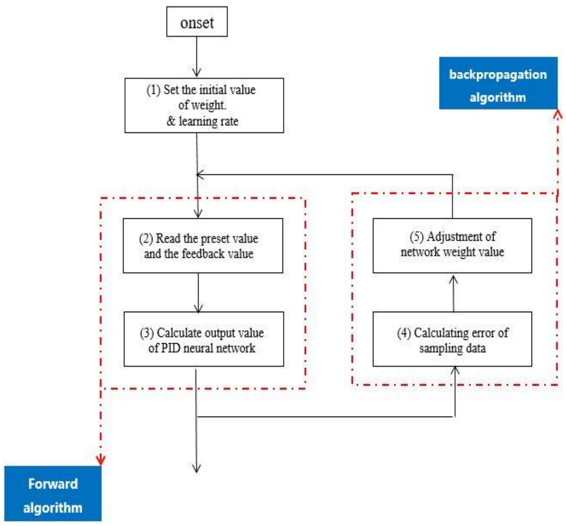


Figure 9: Flow chart of control algorithm

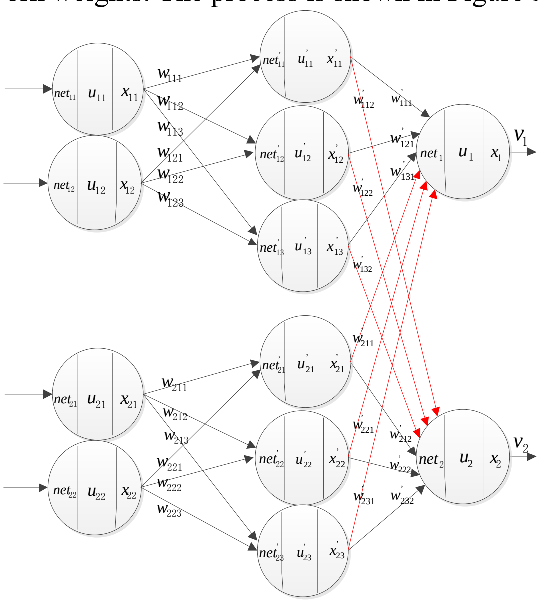


Figure 10: internal structure of MPIDNN controller

4.2.1. forward algorithm. The forward algorithm is as follows. Each neuron is divided into the net (input of the neuron), u (the state function of the neuron), and x (the output of the neuron), as is shown in Figure 10 above:

Input layer: The input layer contains four identical neurons, which receive the Y direction's pre-set value r_1 and feedback y_1, Z direction's pre-set value r_2 and feedback y_2. The input of neurons is:

$$\begin{cases} net_{11}(k) = r_1(k) \\ net_{12}(k) = y_1(k) \\ net_{21}(k) = r_2(k) \\ net_{22}(k) = y_2(k) \end{cases} \tag{15}$$

The output of neurons is as follows and the subscript s is the sequence number of the subnet contained in MPIDNN,s=1, 2. i is the sequence number of neurons in the input layer of each subnet. i=1, 2

$$x_{si}(k)=u_{si}(k) = net_{si}(k) \tag{16}$$

Hidden layer: There are two groups of neurons in this layer. Each group has one proportional neuron, one differential neuron and one integral neuron. They are connected by the weight with the input layer's neurons. Therefore, the input of the hidden neurons is as follows: j is the sequence number of the hidden neurons in the subnet, j=1, 2, 3.

$$net'_{sj}(k) = \sum_{i=1}^{2} w_{sij} x_{si}(k) \tag{17}$$

Each group contains three neurons which respectively have different state function of proportion, integral and differential. Therefore, the output of neurons are as follows:

$$x'_{s1}(k)=u'_{s1}(k) = net'_{s1}(k) \tag{18}$$

$$x'_{s2}(k)=u'_{s2}(k) = u'_{s2}(k-1) + net'_{s2}(k) \tag{19}$$

$$x'_{s3}(k)=u'_{s3}(k) = net'_{s3}(k) - net'_{s3}(k-1) \tag{20}$$

Output layer: There are two neurons in the output layer, which are connected to the 6 neurons in the hidden layer through the weights The input of the neurons is:

$$\begin{cases} net_1(k) = \sum_{i=1}^{2}\sum_{j=1}^{3} w'_{sj1} x'_{sj}(k) \\ net_2(k) = \sum_{i=1}^{2}\sum_{j=1}^{3} w'_{sj2} x'_{sj}(k) \end{cases} \tag{21}$$

The output of neurons is as follows: h is the sequence number of output layer neurons, h=1, 2

$$x_h(k)=u_h(k) = net_h(k) \tag{22}$$

After the above calculation, the output of the entire MPIDNN controller can be obtained as follows:

$$\begin{cases} v_1(k)=x_1(k) \\ v_2(k)=x_2(k) \end{cases} \tag{23}$$

4.2.2. backpropagation algorithm. The aim of the backpropagation algorithm is to modify the connection weights between the two layers by the "error backpropagation method" with the "gradient descent method" as the core, so as to reduce the error. The error is:

$$J = \frac{1}{2}\sum E_h = \frac{1}{2}\sum_{h=1}^{2}\left[r_h(k) - y_h(k)\right]^2 \tag{24}$$

The modifying of weight values between hidden layer and output layer:
According to the weight iteration formula from hidden layer to output layer

$$w'_{sjh}(k+1) = w'_{sjh}(k) - \eta' \frac{\partial J}{\partial w'_{sjh}} \tag{25}$$

In which η' is the learning rate, $\frac{\partial J}{\partial w'_{sjh}}$ can be expressed as:

$$\frac{\partial J}{\partial w'_{sjh}} = \sum_{p=1}^{2}(r_p(k) - y_p(k)) \cdot \mathrm{sgn}\frac{y_p(k+1) - y_p(k)}{v_h(k) - v_h(k-1)} \cdot x'_{sj}(k) \tag{26}$$

The modifying of weight values between input layer and hidden layer:
According to the weight iteration formula from input layer to hidden layer

$$w_{sij}(k+1) = w_{sij}(k) - \eta\frac{\partial J}{\partial w_{sij}} \tag{27}$$

According to the weight iteration formula from input layer to hidden layer:

In which η is the learning rate, $\frac{\partial J}{\partial w_{\mathrm{sij}}}$ can be expressed as:

$$\begin{aligned}\frac{\partial J}{\partial w_{sij}} &= \sum_{p=1}^{2}\sum_{h=1}^{2}(r_p(k) - y_p(k)) \cdot \mathrm{sgn}\frac{y_p(k+1) - y_p(k)}{v_h(k) - v_h(k-1)} \cdot w'_{sjh}(k) \\ &\cdot \mathrm{sgn}\frac{u'_{sj}(k) - u'_{sj}(k-1)}{net'_{sj}(k) - net'_{sj}(k-1)} \cdot x_{si}(k)\end{aligned} \tag{28}$$

5. Simulation result analysis

In the process of simulation, the predetermined three-dimensional trajectory is synthesized with two sinusoidal curves in the horizontal and vertical direction. The starting point coordinates of the simulation starting point are (0, 0, 800m), and the initial velocity is 10m/s. The pre-set trajectory is:

$$\begin{cases} x = x \\ y = 500 \cdot \sin(0.002 \cdot x) \\ z = 100 \cdot \sin(0.002 \cdot x) + 800 \end{cases} \tag{29}$$

The simulation results are shown in Figure 11 as follows:

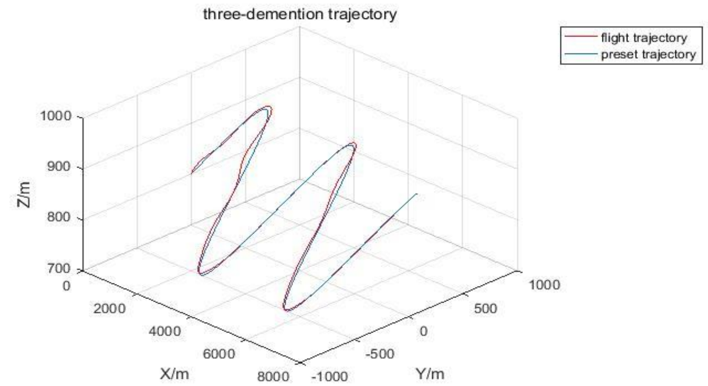


Figure 11: the comparation between flight trajectory and pre-set trajectory

According to the simulation results, during the flight, the UPP can fly according to the pre-set trajectory. The flight control of the trajectory is realized. To better analyse the result of yaw control and longitudinal control, the projections of the three-dimensional trajectory on the horizontal and vertical planes are analysed respectively.

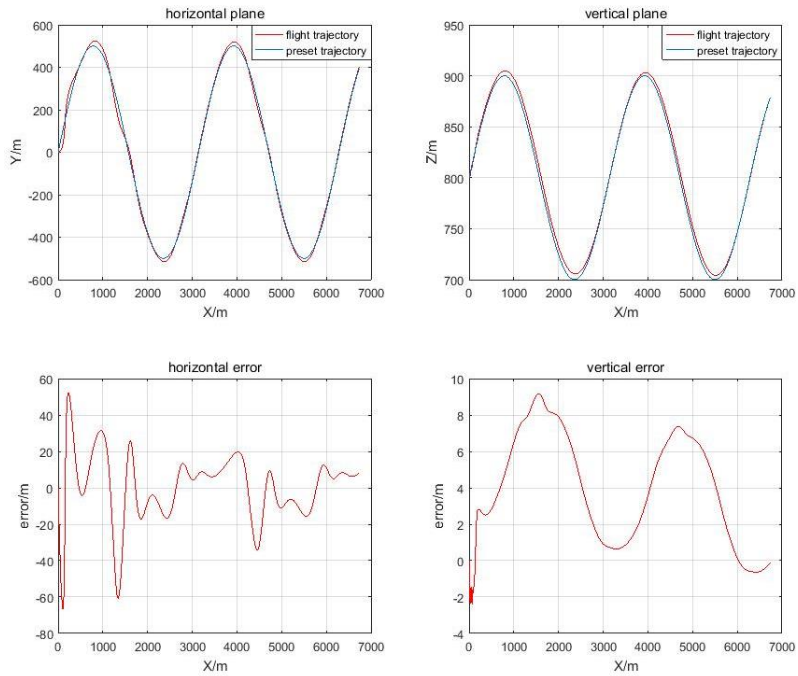


Figure 12: the comparation of trajectories and error in horizontal and vertical plane

As can be seen from the Figure 12 the maximum error is less than 25m when the UPP begins to stay in a stable flight state in the horizontal direction. In the vertical direction, the UPP flies well in accordance with the pre-set trajectory, and the maximum error can remain within 10 m.

Also, the values of the control variables of the PIDNN controller are recorded as follows:

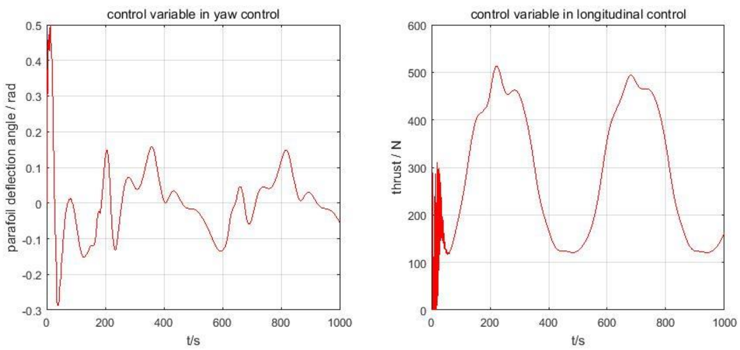


Figure 13: the value of control variable changing with time

As can be seen from Figure 13, the value of the yaw control variable is within 0.2 rad after being stable, and the turning radius is within the required range. The control range in the vertical direction is 0-500N, which is also within the effective thrust range of the propeller which can prevent the stall.

6. Conclusion

The key factors in the flight process are the 6 DOF model of the UPP, the structure of the PIDNN and the design of the control algorithm. From the simulation results, it can be seen that the PIDNN controller can output the control variables satisfying the control characteristics of the UPP, and control the UPP to fly according to the predetermined trajectory with a good fit.

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